



Cluster Analysis I

Lecture 11

肖升生 博士

Dr. Shengsheng Xiao

Email: Xiao.shengsheng@shufe.edu.cn



Outline

- What is cluster analysis
- Similarity and density measurement
- Types of cluster
- K-means and its variant model
- Case study

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Supervised Learning

Feature Space $\mathcal{X}$



Words in a document

Label Space $\mathcal{Y}$

“Sports”
“News”
“Science”
...



Share Price
“\$ 24.50”

Task: Given $X \in \mathcal{X}$, predict $Y \in \mathcal{Y}$.

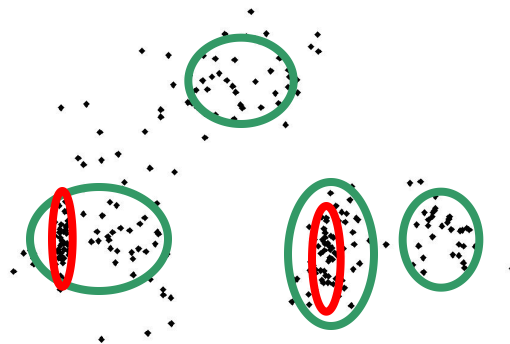
Unsupervised Learning

“Learning from unlabeled/unannotated data” (without supervision)



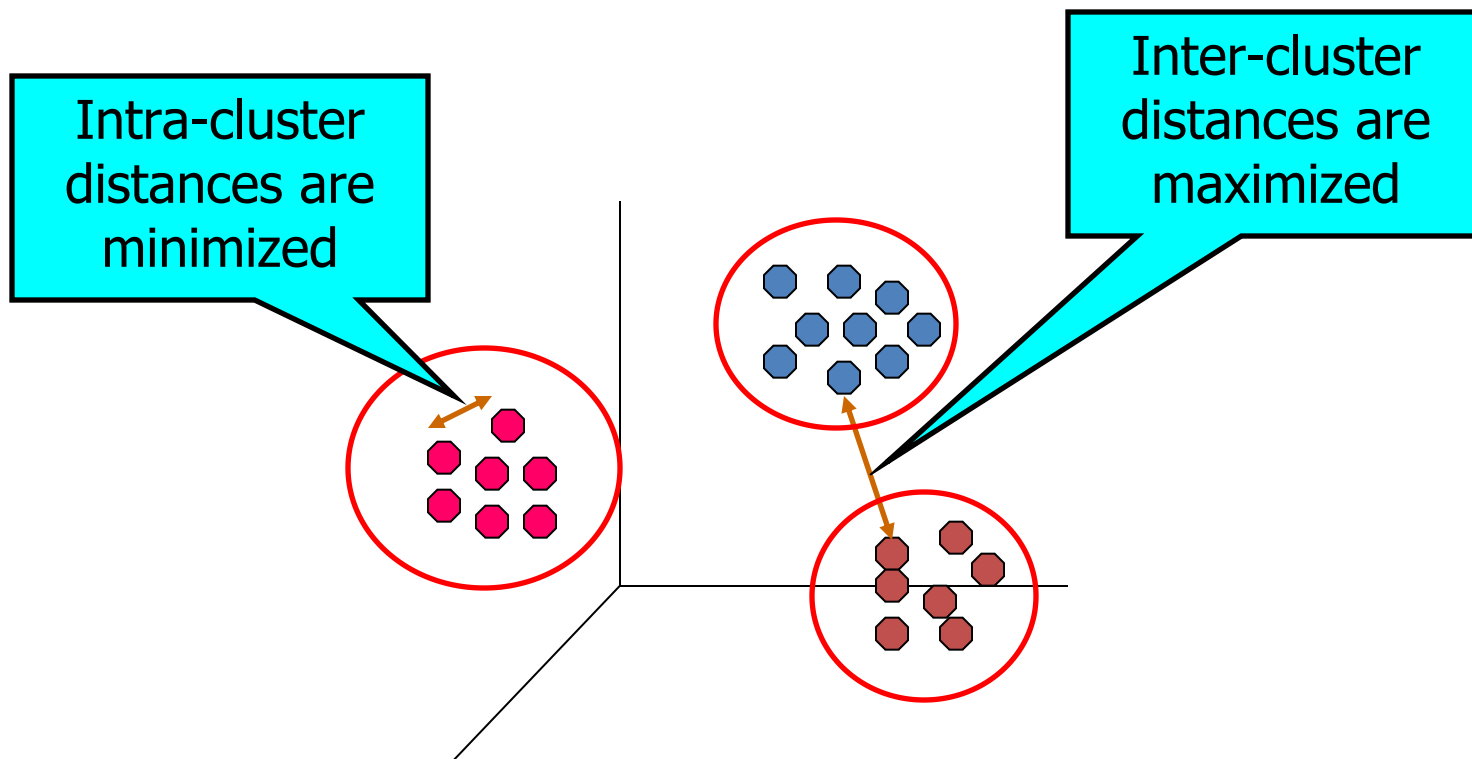
What can we predict/ describe from unlabeled data?

- Groups or clusters in the data

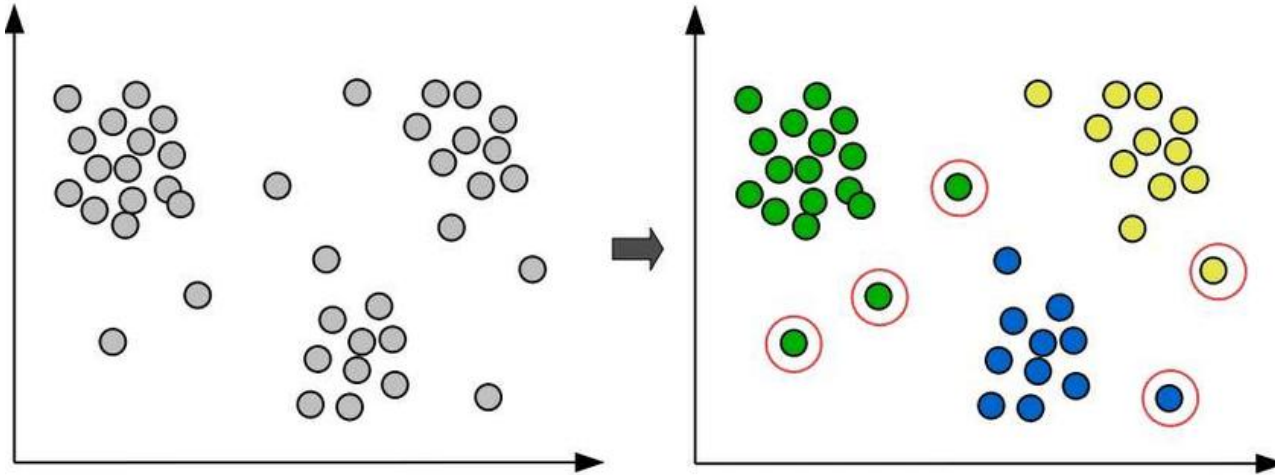


What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



Applications of Cluster Analysis



- **Understanding**

- Group related documents for browsing,
- group genes and proteins that have similar functionality, or
- group stocks with similar price fluctuations

- **Summarization**

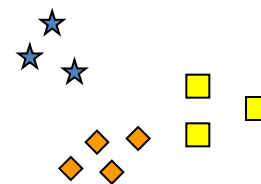
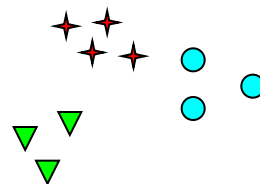
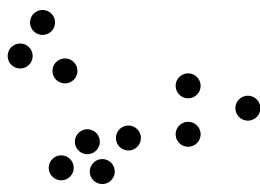
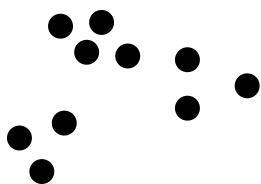
- Reduce the size of large data sets



What is not Cluster Analysis?

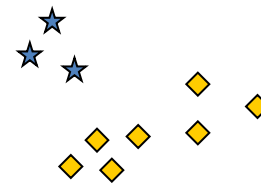
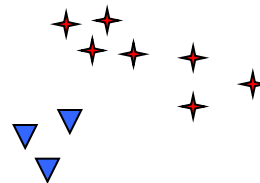
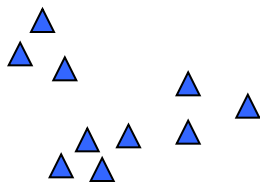
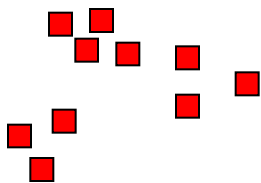
- Supervised classification
 - Have class label information
- Simple segmentation
 - Dividing students into different registration groups alphabetically, by last name
- Results of a query
 - Groupings are a result of an external specification
- Graph partitioning
 - Some mutual relevance and synergy, but areas are not identical

Cluster can be Ambiguous



How many clusters?

Six Clusters

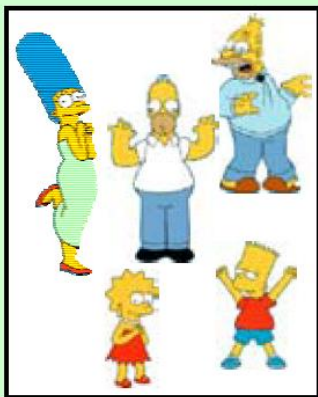


Two Clusters

Four Clusters

Cluster can be Ambiguous

Clustering is subjective



Simpson's Family



School Employees



Females



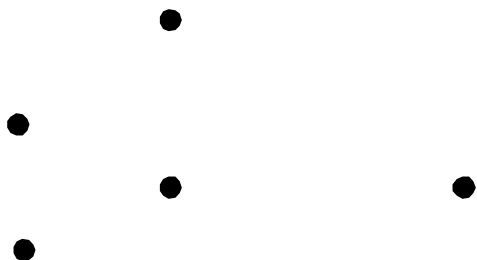
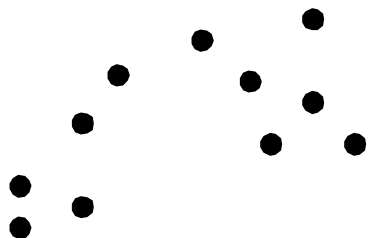
Males



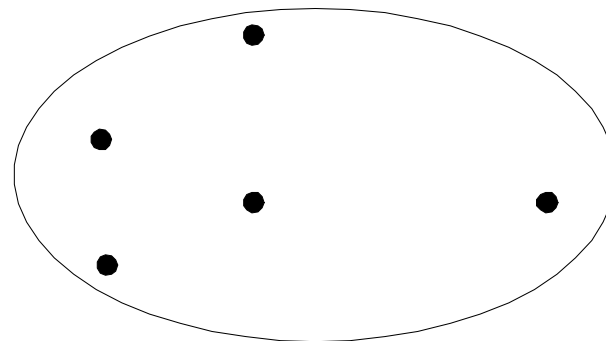
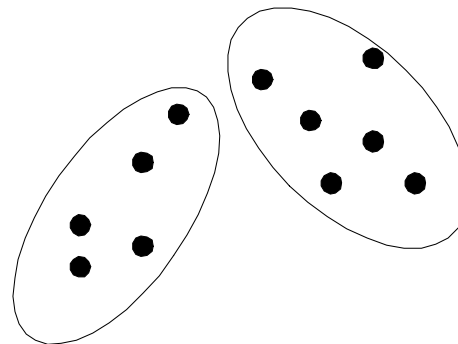
Types of Clusterings

- A **clustering** is a set of clusters
- Partitional Clustering
 - A division data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset
- Hierarchical clustering
 - A set of nested clusters organized as a hierarchical tree

Partitional Clustering

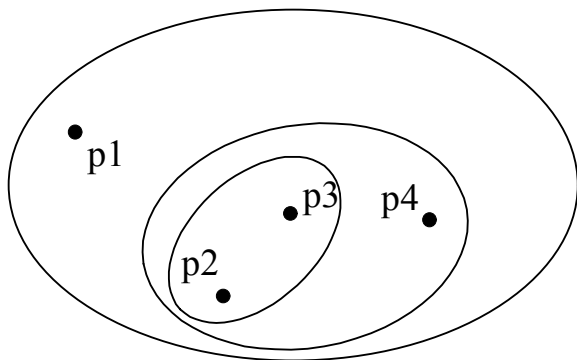


Original Points

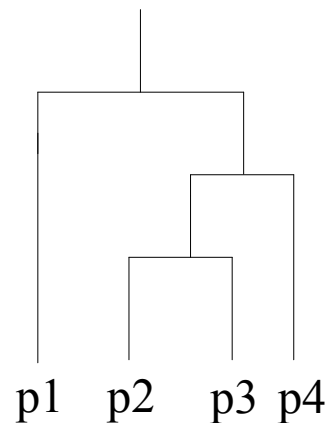


A Partitional Clustering

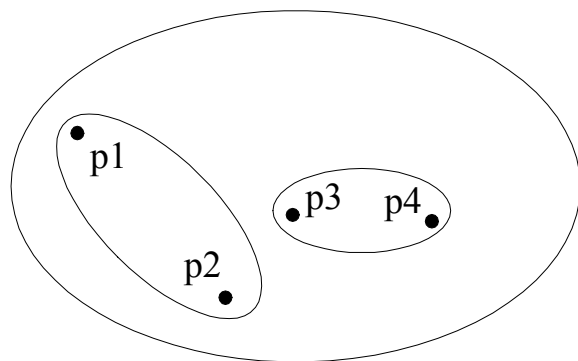
Hierarchical Clustering



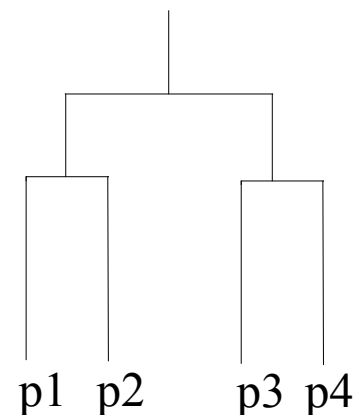
Traditional Hierarchical Clustering



Traditional Dendrogram



Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

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- Types of cluster
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What is Similarity?



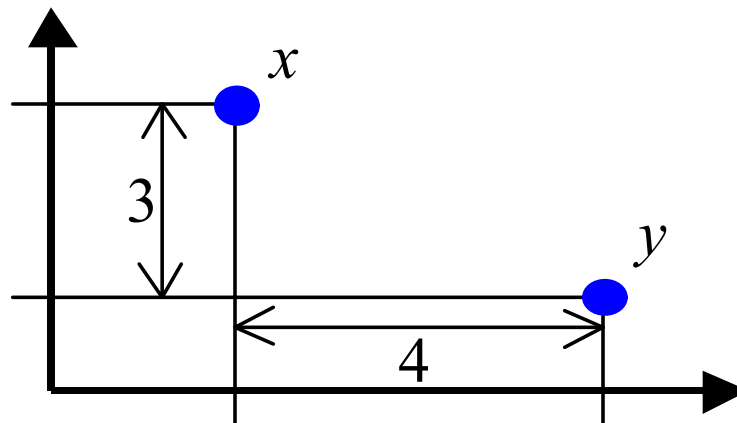
Hard to define!
But we *know it*
when we see it

- The real meaning of similarity is a philosophical question.
- We will use distance between vectors to measure the similarity

Distance metrics

$$x = (x_1, x_2, \dots, x_p)$$

$$y = (y_1, y_2, \dots, y_p)$$



Euclidean distance

$$d(x, y) = \sqrt{\sum_{i=1}^p |x_i - y_i|^2}$$

5

Manhattan distance

$$d(x, y) = \sum_{i=1}^p |x_i - y_i|$$

7

Sup-distance

$$d(x, y) = \max_{1 \leq i \leq p} |x_i - y_i|$$

4

Minkowski Distance

- Minkowski Distance is a generalization of Euclidean Distance

$$\text{dist} = \left(\sum_{k=1}^n |p_k - q_k|^r \right)^{\frac{1}{r}}$$

Where r is a parameter, n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k th attributes (components) or data objects p and q .

Minkowski Distance: Examples

- $r = 1$. City block (Manhattan, taxicab, L_1 norm) distance.
 - A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors
- $r = 2$. Euclidean distance
- $r \rightarrow \infty$. “supremum” ($L_{\max}$ norm, L_{∞} norm) distance.
 - This is the maximum difference between any component of the vectors
- Do not confuse r with n , i.e., all these distances are defined for all numbers of dimensions.

Minkowski Distance

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_{∞}	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

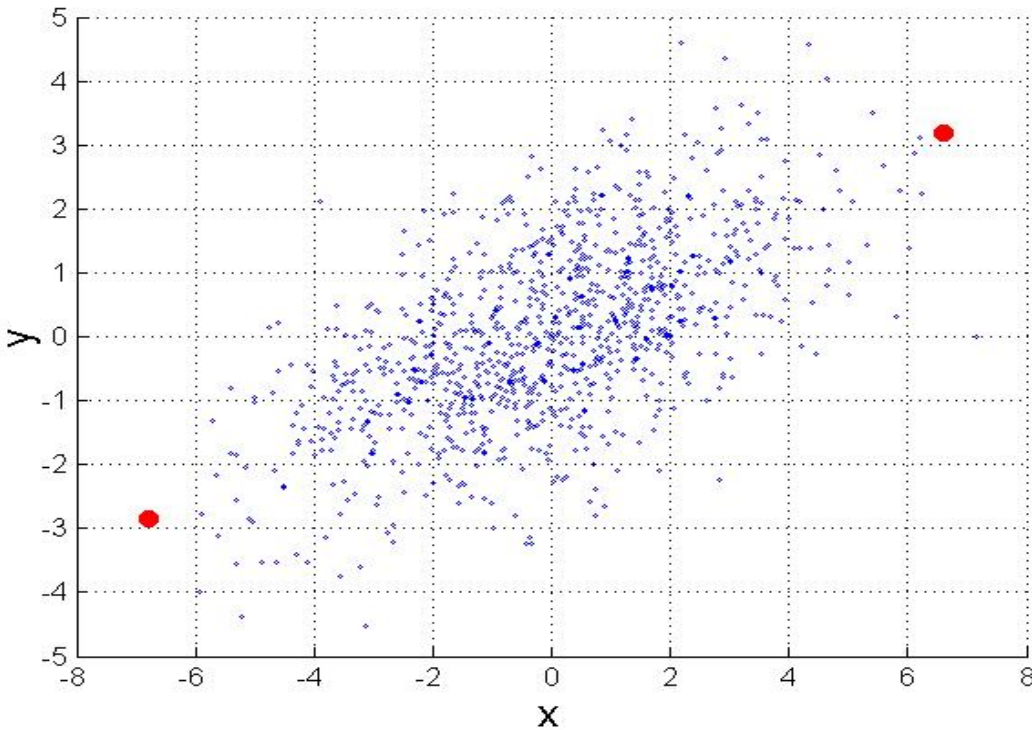
Distance Matrix

Mahalanobis Distance

$$\text{mahalanobis } s(p, q) = (p - q) \Sigma^{-1} (p - q)^T$$

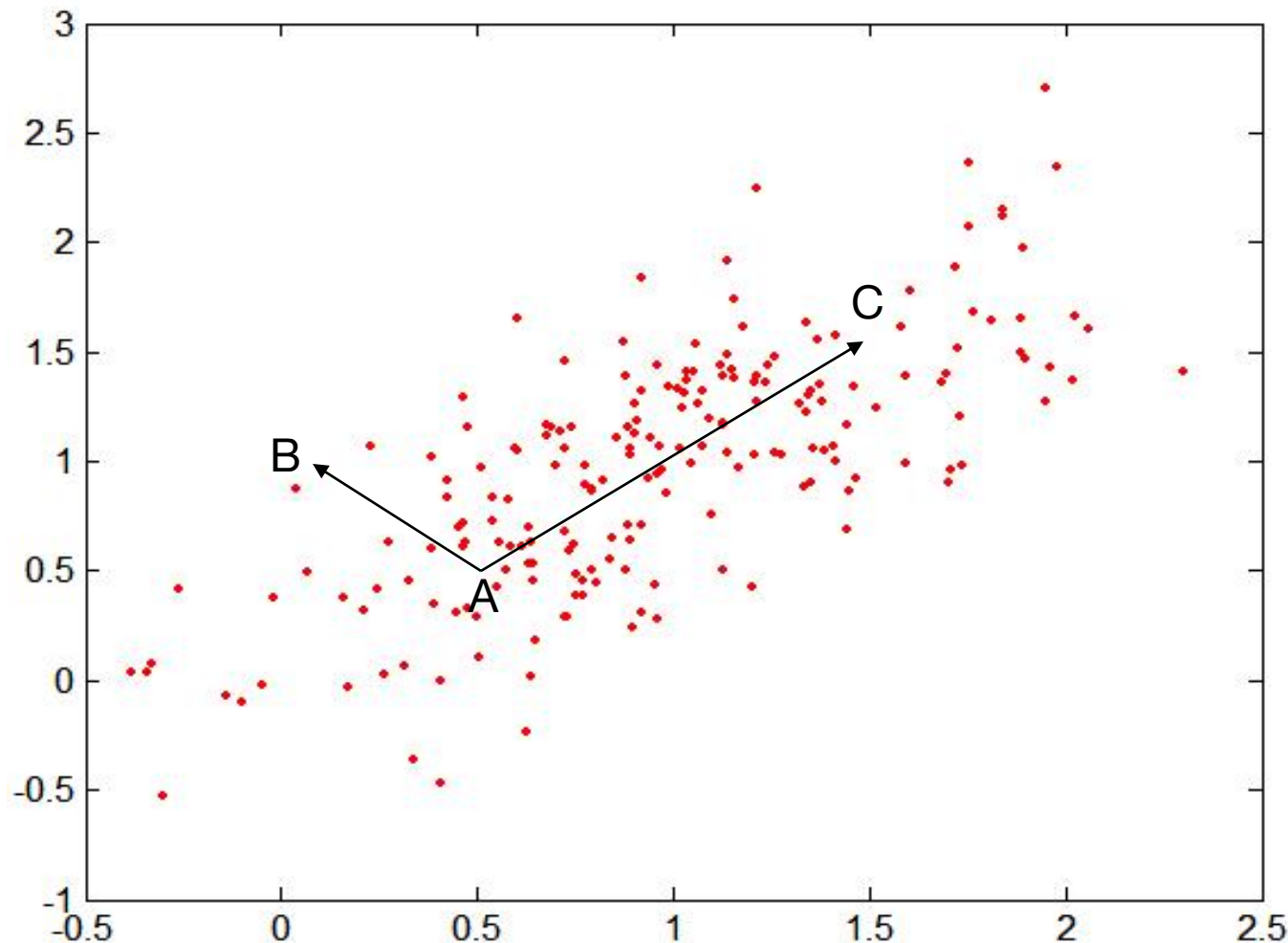
Σ is the covariance matrix of the input data X

$$\Sigma_{j,k} = \frac{1}{n-1} \sum_{i=1}^n (X_{ij} - \bar{X}_j)(X_{ik} - \bar{X}_k)$$



For red points, the Euclidean distance is 14.7, Mahalanobis distance is 6.

Mahalanobis Distance



Covariance Matrix:

$$\Sigma = \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}$$

A: (0.5, 0.5)

B: (0, 1)

C: (1.5, 1.5)

Mahal(A,B) = 5

Mahal(A,C) = 4

Common Properties of a Distance

- Distances, such as the Euclidean distance, have some well known properties.
 1. $d(p, q) \geq 0$ for all p and q and $d(p, q) = 0$ only if $p = q$. (Positive definiteness)
 2. $d(p, q) = d(q, p)$ for all p and q . (Symmetry)
 3. $d(p, r) \leq d(p, q) + d(q, r)$ for all points p, q , and r . (Triangle Inequality)

where $d(p, q)$ is the distance (dissimilarity) between points (data objects), p and q .
- A distance that satisfies these properties is a **metric**

Similarity Between Binary Vectors

- Common situation is that objects, p and q , have only binary attributes

- Compute similarities using the following quantities

M_{01} = the number of attributes where p was 0 and q was 1

M_{10} = the number of attributes where p was 1 and q was 0

M_{00} = the number of attributes where p was 0 and q was 0

M_{11} = the number of attributes where p was 1 and q was 1

- Simple Matching and Jaccard Coefficients

SMC = number of matches / number of attributes

$$= (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00})$$

JC = number of 11 matches / number of not-both-zero attributes values

$$= (M_{11}) / (M_{01} + M_{10} + M_{11})$$

SMC versus Jaccard: Example

$$p = 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0$$

$$q = 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 1$$

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)

$M_{10} = 1$ (the number of attributes where p was 1 and q was 0)

$M_{00} = 7$ (the number of attributes where p was 0 and q was 0)

$M_{11} = 0$ (the number of attributes where p was 1 and q was 1)

$$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) = (0+7) / (2+1+0+7) = 0.7$$

$$JC = (M_{11}) / (M_{01} + M_{10} + M_{11}) = 0 / (2 + 1 + 0) = 0$$

Cosine Similarity

- If d_1 and d_2 are two document vectors, then

$$\cos(d_1, d_2) = (d_1 \bullet d_2) / \|d_1\| \|d_2\| ,$$

where $\bullet$ indicates vector dot product and $\|d\|$ is the length of vector d .

- Example:

$$d_1 = 3 \ 2 \ 0 \ 5 \ 0 \ 0 \ 0 \ 2 \ 0 \ 0$$

$$d_2 = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 2$$

$$d_1 \bullet d_2 = 3*1 + 2*0 + 0*0 + 5*0 + 0*0 + 0*0 + 0*0 + 2*1 + 0*0 + 0*2 = 5$$

$$\|d_1\| = (3*3 + 2*2 + 0*0 + 5*5 + 0*0 + 0*0 + 0*0 + 2*2 + 0*0 + 0*0)^{0.5} = (42)^{0.5} = 6.481$$

$$\|d_2\| = (1*1 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 1*1 + 0*0 + 2*2)^{0.5} = (6)^{0.5} = 2.245$$

$$\cos(d_1, d_2) = .3150$$

Extended Jaccard Coefficient

- Variation of Jaccard for continuous or count attributes
 - Reduces to Jaccard for binary attributes

$$T(p, q) = \frac{p \bullet q}{\|p\|^2 + \|q\|^2 - p \bullet q}$$

Correlation

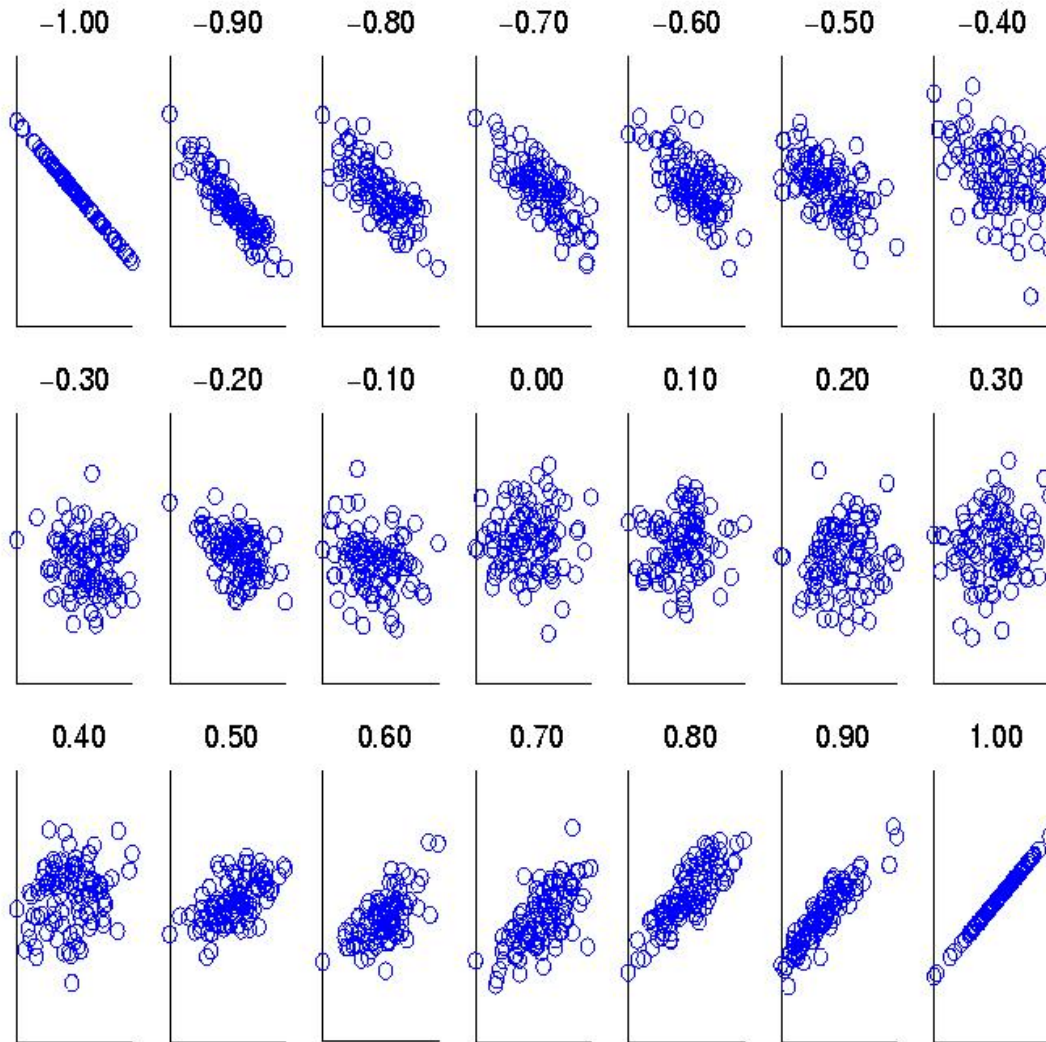
- Correlation measures the linear relationship between objects
- To compute correlation, we standardize data objects, p and q , and then take their dot product

$$p'_k = (p_k - \text{mean}(p)) / \text{std}(p)$$

$$q'_k = (q_k - \text{mean}(q)) / \text{std}(q)$$

$$\text{correlation}(p, q) = p' \bullet q'$$

Visually Evaluating Correlation



Scatter plots showing the similarity from -1 to 1 .

General Approach for Combining Similarities

- Sometimes attributes are of many different types, but an overall similarity is needed.

1. For the k^{th} attribute, compute a similarity, s_k , in the range $[0, 1]$.
2. Define an indicator variable, δ_k , for the k^{th} attribute as follows:

$$\delta_k = \begin{cases} 0 & \text{if the } k^{th} \text{ attribute is a binary asymmetric attribute and both objects have} \\ & \text{a value of 0, or if one of the objects has a missing values for the } k^{th} \text{ attribute} \\ 1 & \text{otherwise} \end{cases}$$

3. Compute the overall similarity between the two objects using the following formula:

$$similarity(p, q) = \frac{\sum_{k=1}^n \delta_k s_k}{\sum_{k=1}^n \delta_k}$$

Using Weights to Combine Similarities

- May not want to treat all attributes the same.
 - Use weights w_k which are between 0 and 1 and sum to 1.

$$\text{similarity}(p, q) = \frac{\sum_{k=1}^n w_k \delta_k s_k}{\sum_{k=1}^n \delta_k}$$

$$\text{distance}(p, q) = \left(\sum_{k=1}^n w_k |p_k - q_k|^r \right)^{1/r}$$

- Density-based clustering require a notion of density
- Examples:
 - Euclidean density
 - Euclidean density = number of points per unit volume
 - Graph-based density

Euclidean Density – Cell-based

- Simplest approach is to divide region into a number of rectangular cells of equal volume and define density as # of points

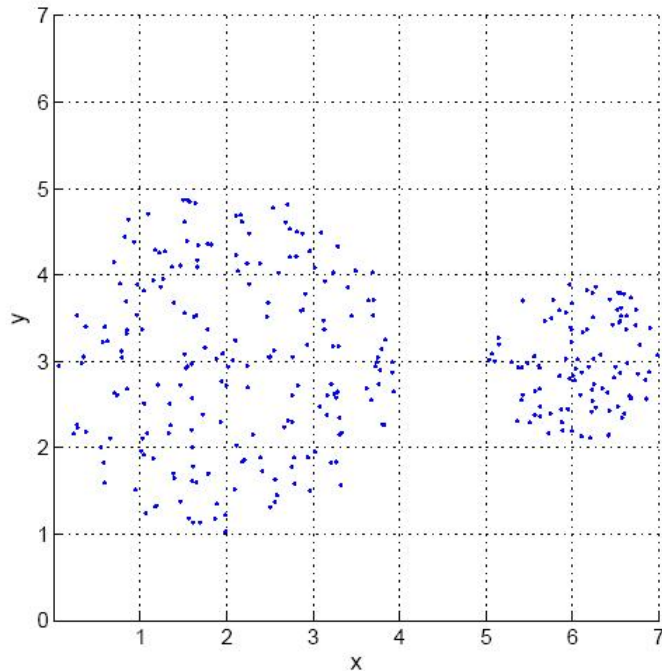


Figure 7.13. Cell-based density.

0	0	0	0	0	0	0
0	0	0	0	0	0	0
4	17	18	6	0	0	0
14	14	13	13	0	18	27
11	18	10	21	0	24	31
3	20	14	4	0	0	0
0	0	0	0	0	0	0

Table 7.6. Point counts for each grid cell.

Euclidean Density – Center-based

- Euclidean density is the number of points within a specified radius of the point

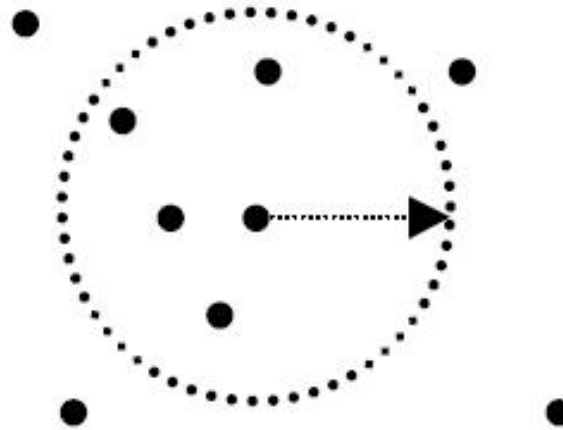


Figure 7.14. Illustration of center-based density.

- What is cluster analysis
- Similarity and density measurement
- Types of cluster
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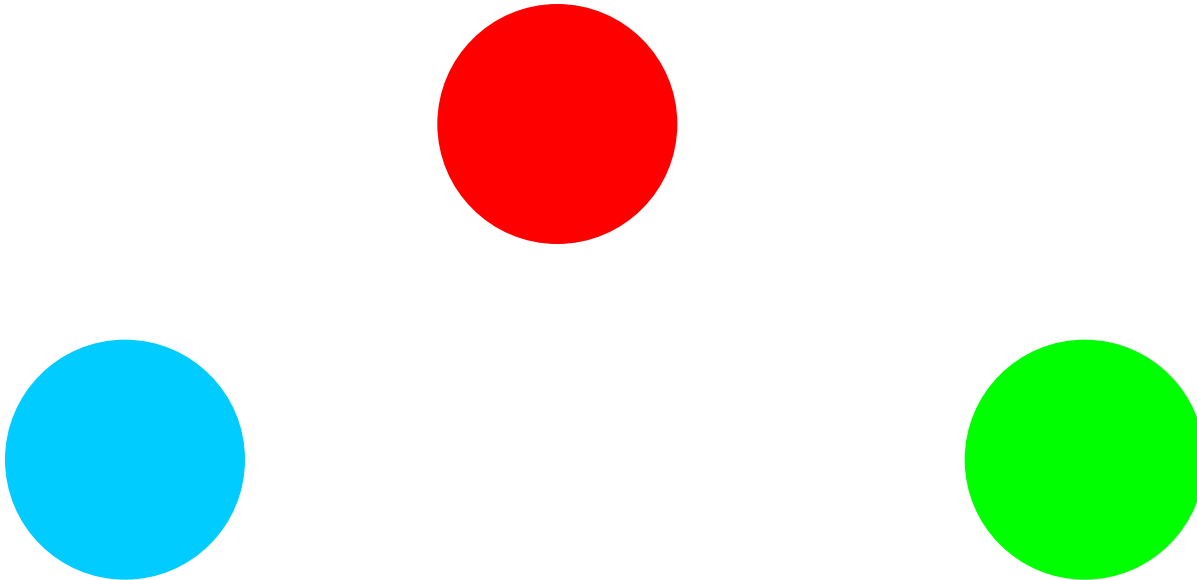


Types of Clusters

- **Well-separated clusters**
- **Center-based clusters**
- **Contiguous clusters**
- **Density-based clusters**
- **Property or Conceptual**
- **Described by an Objective Function**

Types of Clusters: Well-Separated

- Well-Separated Clusters:
 - A cluster is a set of points such that any point in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.

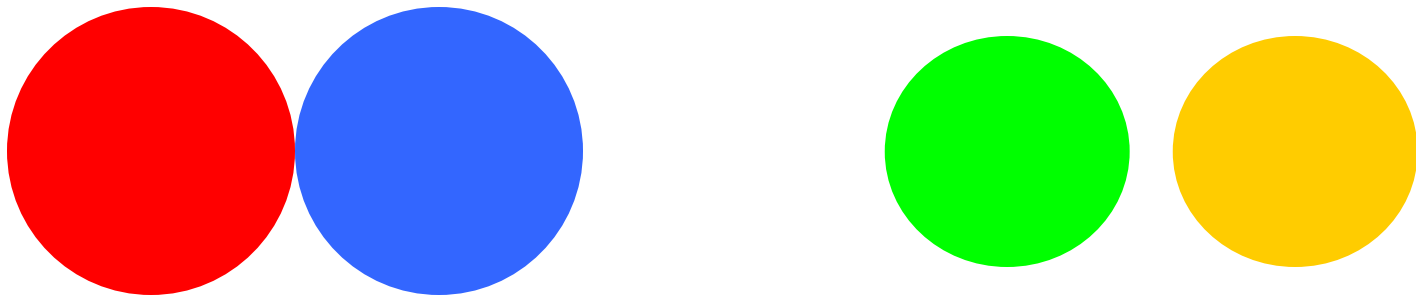


3 well-separated clusters

Types of Clusters: Center-Based

- Center-based

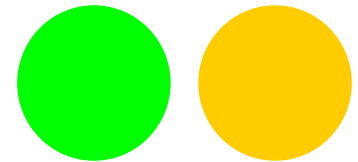
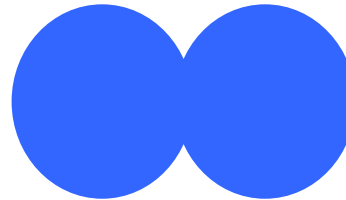
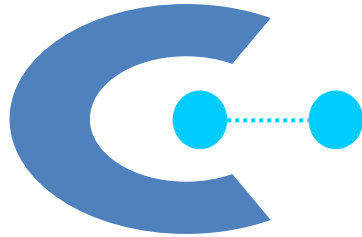
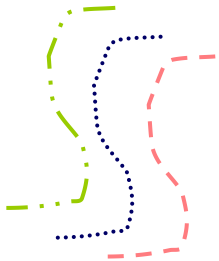
- A cluster is a set of objects such that an object in a cluster is closer (more similar) to the “center” of a cluster, than to the center of any other cluster
- The center of a cluster is often a **centroid**, the average of all the points in the cluster, or a **medoid**, the most “representative” point of a cluster



4 center-based clusters

Types of Clusters: Contiguity-Based

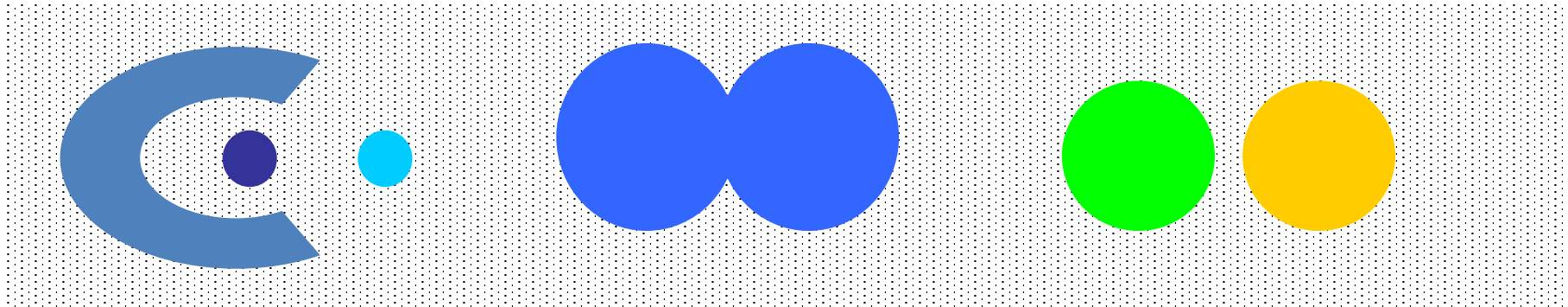
- Contiguous Cluster (Nearest neighbor or Transitive)
 - A cluster is a set of points such that a point in a cluster is closer (or more similar) to one or more other points in the cluster than to any point not in the cluster.



8 contiguous clusters

Types of Clusters: Density-Based

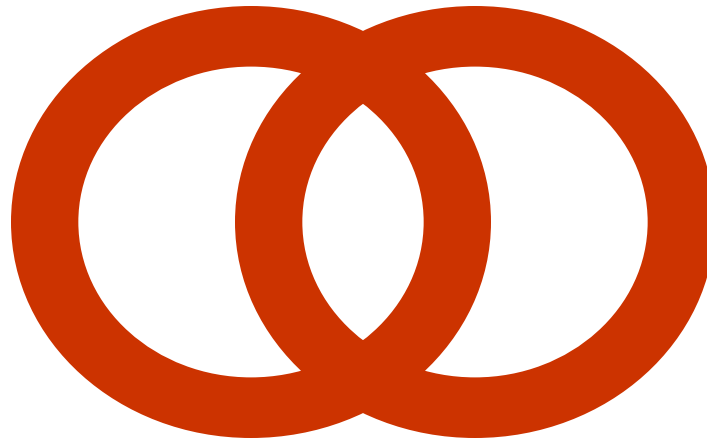
- Density-based
 - A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
 - Used when the clusters are irregular or intertwined, and when noise and outliers are present.



6 density-based clusters

Types of Clusters: Conceptual Clusters

- Shared Property or Conceptual Clusters
 - Finds clusters that share some common property or represent a particular concept.



2 Overlapping Circles

- Clusters Defined by an Objective Function
 - Finds clusters that minimize or maximize an objective function
 - Enumerate all possible ways of dividing the points into clusters and evaluate the 'goodness' of each potential set of clusters by using the given objective function.
 - Can have **global or local objectives**.
 - Hierarchical clustering algorithms typically have local objectives
 - Partitional algorithms typically have global objectives
 - A variation of the global objective function approach is to **fit the data to a parameterized model**
 - Mixture models : The data is a 'mixture' of a number of statistical distributions
 - Parameters for the model are determined from the data

Clustering Strategy

- Map the clustering problem to a different domain and solve a related problem in that domain
 - Proximity matrix defines a weighted graph, where the nodes are the points being clustered, and the weighted edges represent the proximities between points
 - Clustering is equivalent to breaking the graph into connected components, one for each cluster.
 - Want to minimize the edge weight between clusters and maximize the edge weight within clusters



Clustering Algorithms

- K-means and its variants
- Hierarchical clustering
- Density-based clustering

Outline

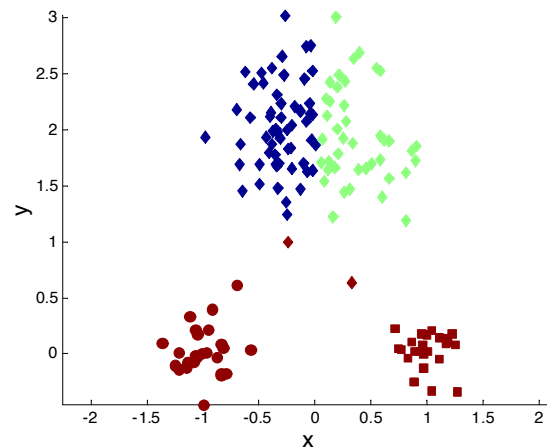
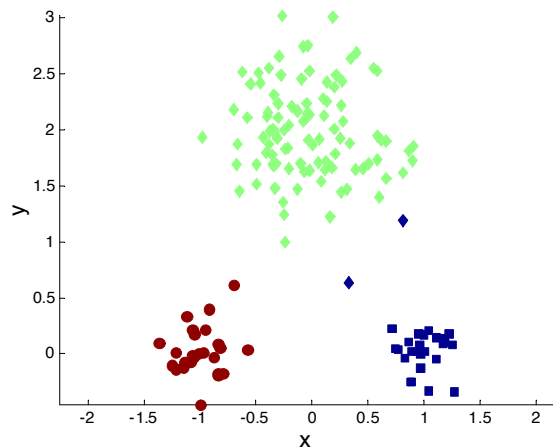
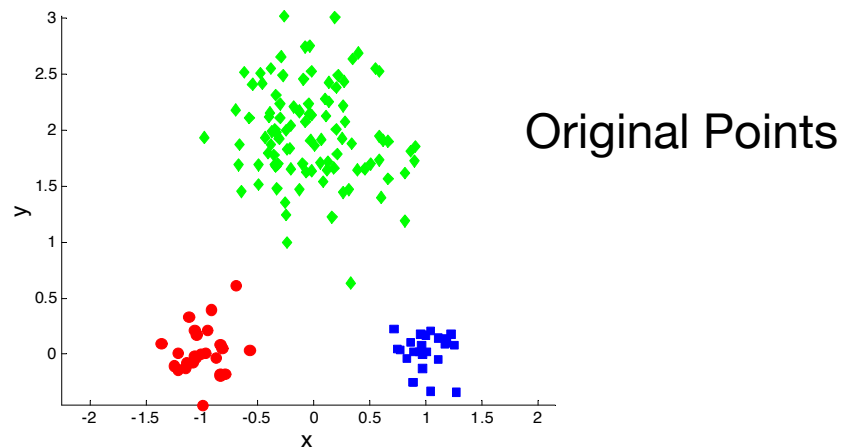
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K-means Clustering

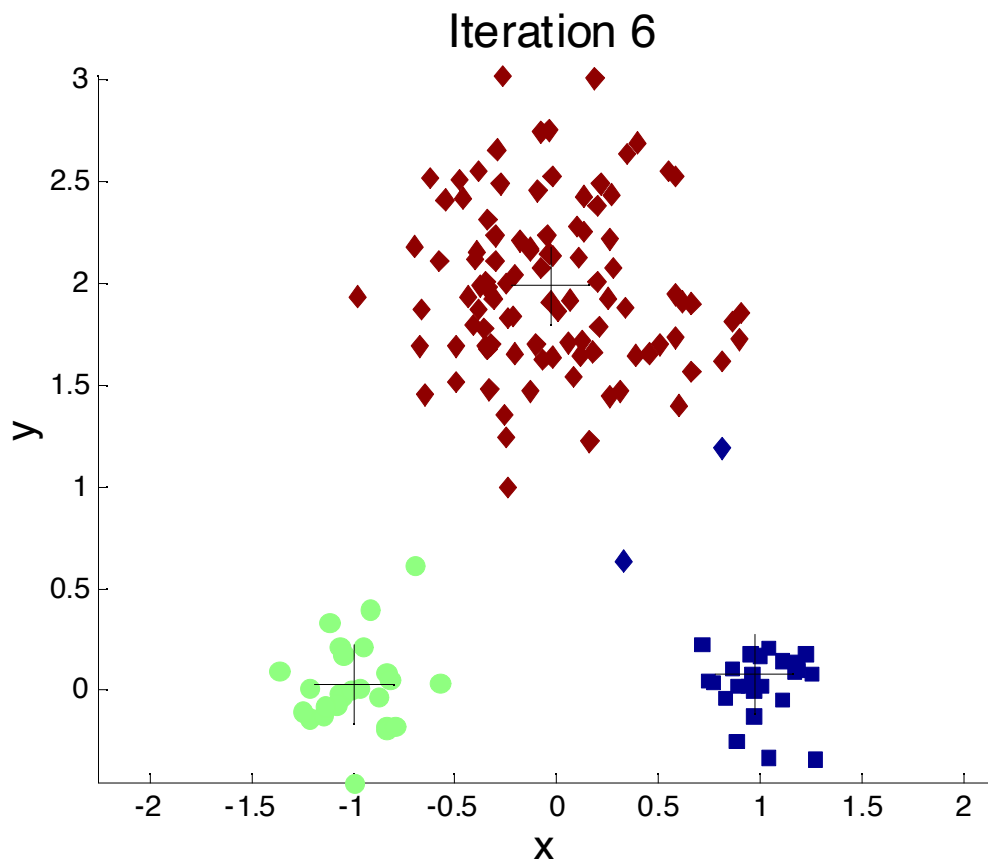
- Partitional clustering approach
- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid
- Number of clusters, K , must be specified
- The basic algorithm is very simple

-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

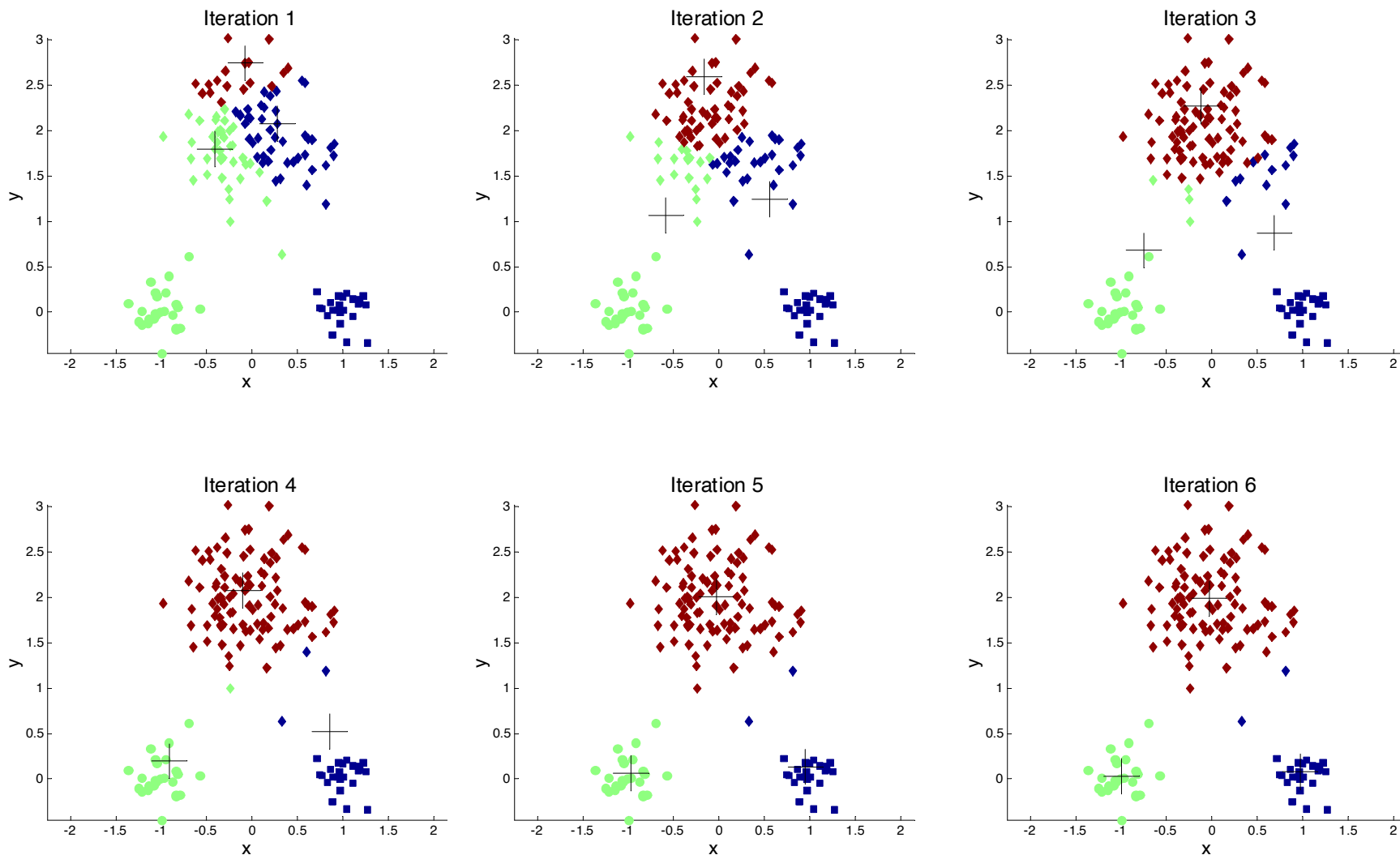
Two different K-means Clusterings



Importance of Choosing Initial Centroids



Importance of Choosing Initial Centroids



K-means Clustering – Details

- Initial centroids are often chosen randomly.
 - Clusters produced vary from one run to another.
- The centroid is (typically) the mean of the points in the cluster.
- ‘Closeness’ is measured by **Euclidean distance, cosine similarity, correlation, etc.**
- K-means will converge for common similarity measures mentioned above.
- Most of the convergence happens in the first few iterations.
 - Often the stopping condition is changed to ‘Until relatively few points change clusters’
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters, I = number of iterations, d = number of attributes

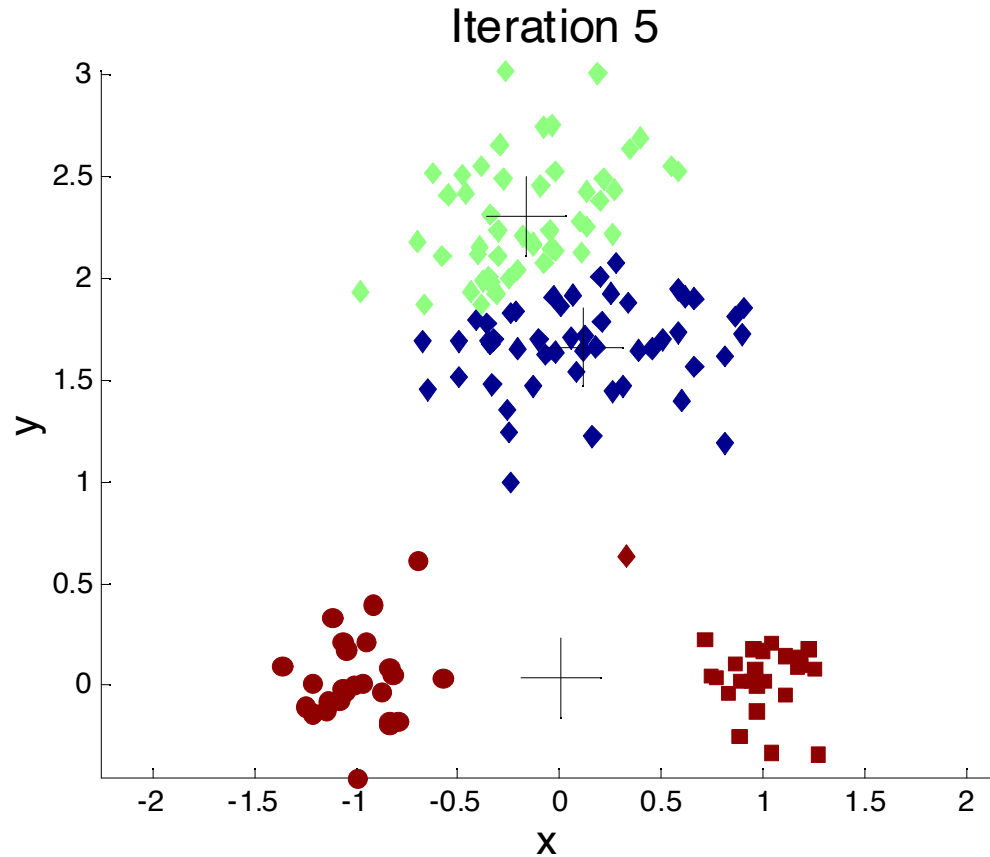
Evaluating K-means Clusters

- Most common measure is Sum of Squared Error (SSE)
 - For each point, the error is the distance to the nearest cluster
 - To get SSE, we square these errors and sum them.

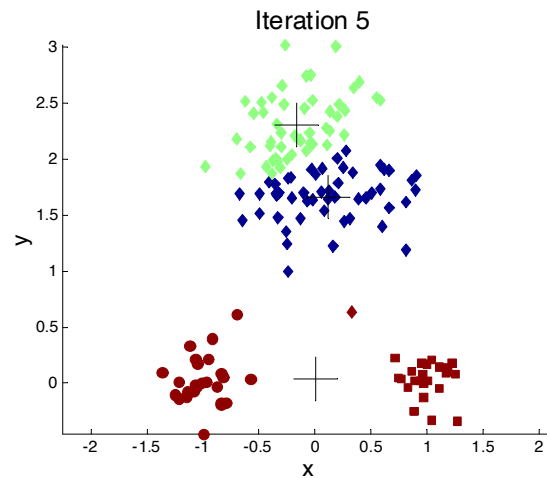
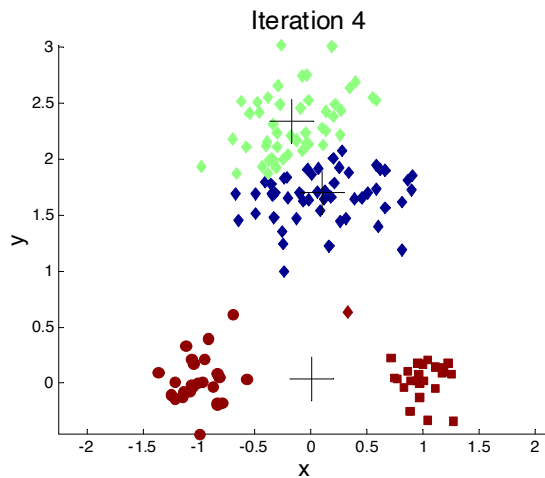
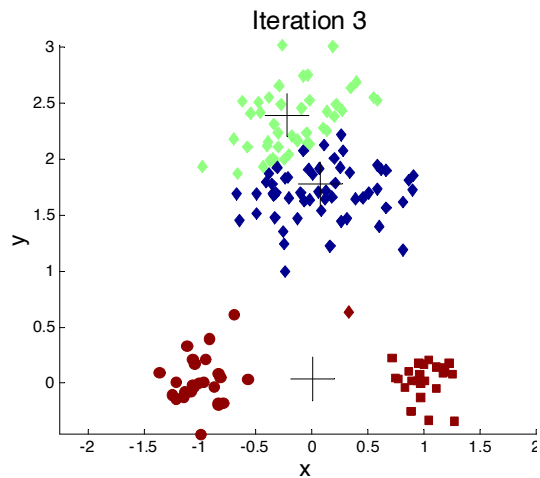
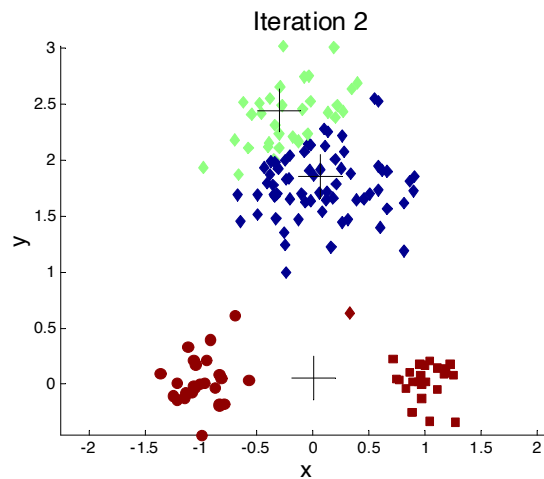
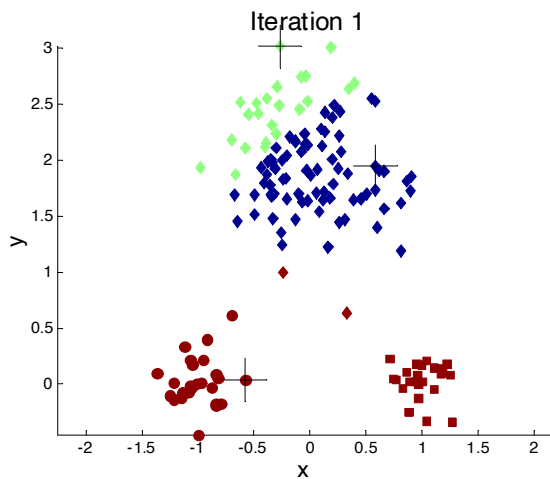
$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

- x is a data point in cluster C_i and m_i is the representative point for cluster C_i
 - can show that m_i corresponds to the center (mean) of the cluster
- Given two clusters, we can choose the one with the smallest error
- One easy way to reduce SSE is to increase K , the number of clusters
 - A good clustering with smaller K can have a lower SSE than a poor clustering with higher K

Importance of Choosing Initial Centroids ...



Importance of Choosing Initial Centroids ...



Solutions to Initial Centroids Problem

- Multiple runs
 - Helps, but probability is not on your side
- Sample and use hierarchical clustering to determine initial centroids
- Select more than k initial centroids and then select among these initial centroids
 - Select most widely separated
- Pre-processing and Post-processing
- Bisecting K-means
 - Not as susceptible to initialization issues

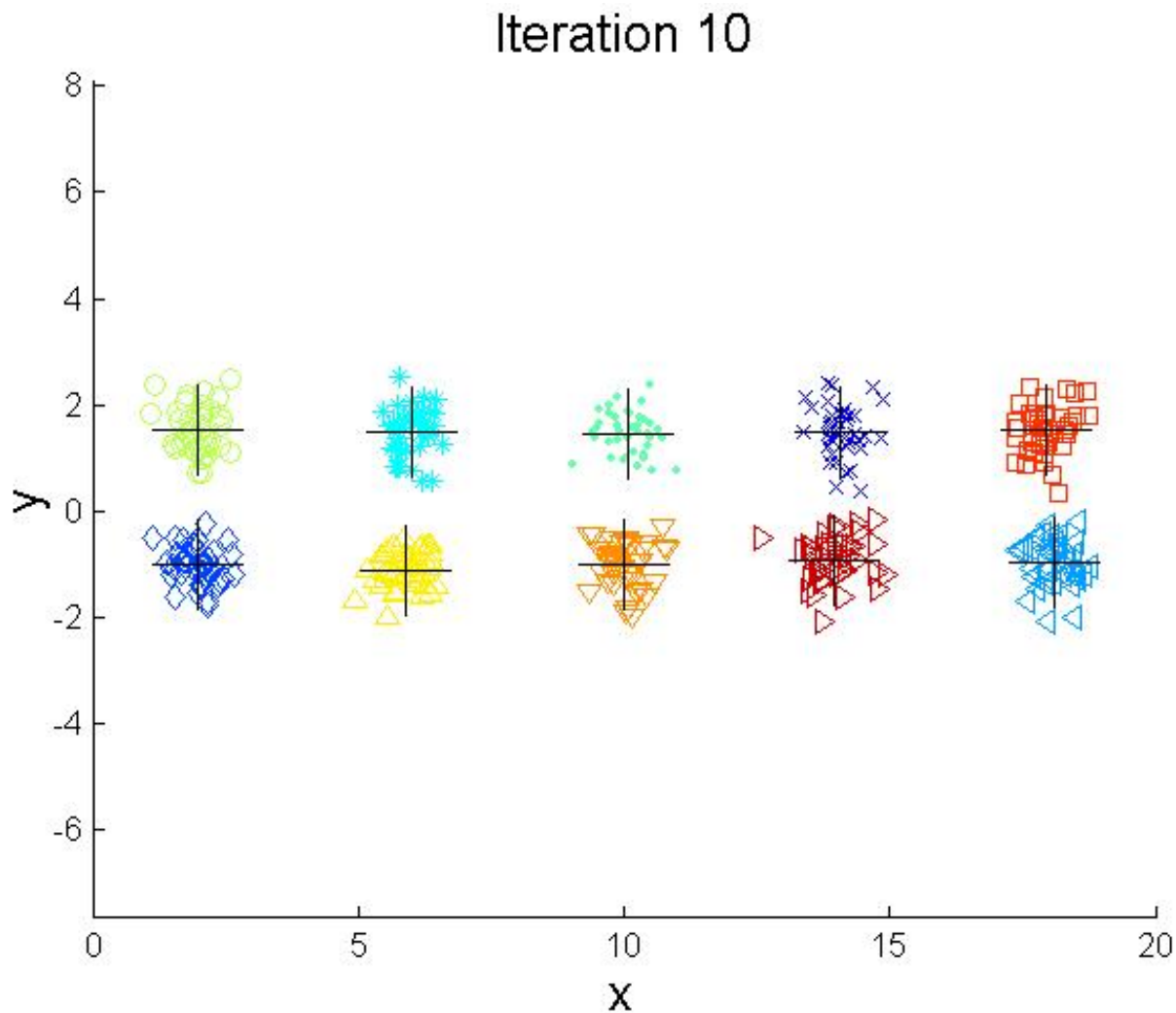
Pre-processing and Post-processing

- Pre-processing
 - Normalize the data
 - Eliminate outliers
- Post-processing
 - Eliminate small clusters that may represent outliers
 - Split ‘loose’ clusters, i.e., clusters with relatively high SSE
 - Merge clusters that are ‘close’ and that have relatively low SSE
 - Can use these steps during the clustering process

- Bisecting K-means algorithm
 - Variant of K-means that can produce a partitional or a hierarchical clustering

```
1: Initialize the list of clusters to contain the cluster containing all points.  
2: repeat  
3:   Select a cluster from the list of clusters  
4:   for  $i = 1$  to number_of_iterations do  
5:     Bisect the selected cluster using basic K-means  
6:   end for  
7:   Add the two clusters from the bisection with the lowest SSE to the list of clusters.  
8: until Until the list of clusters contains  $K$  clusters
```

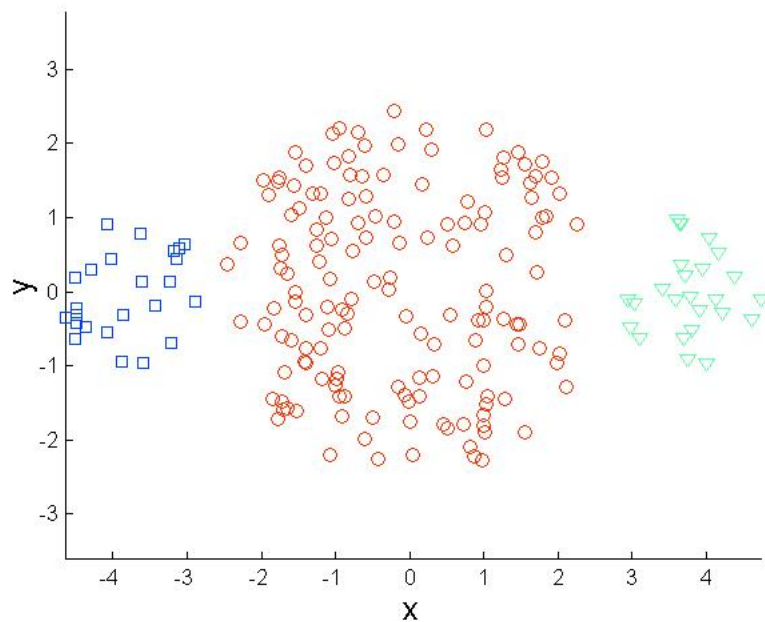
Bisecting K-means Example



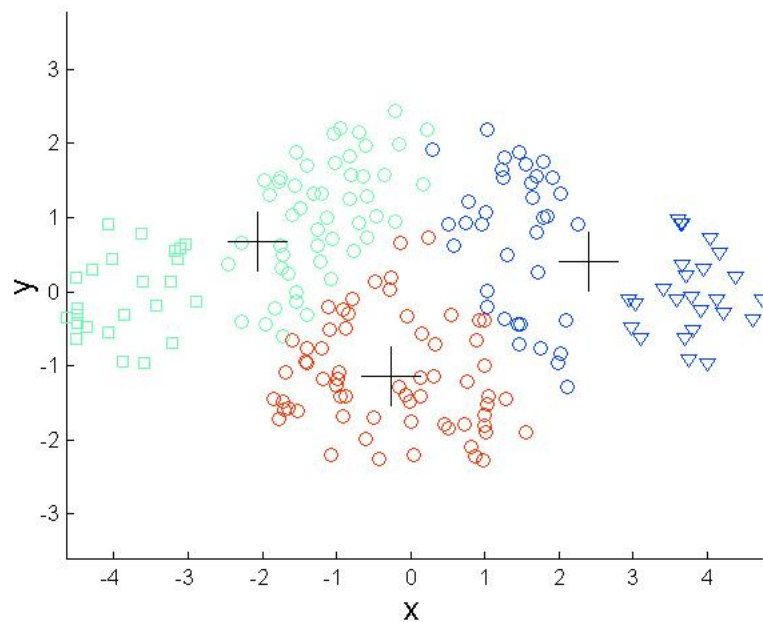
Limitations of K-means

- K-means has problems when clusters are of differing
 - Sizes
 - Densities
 - Non-globular shapes
- K-means has problems when the data contains outliers.

Limitations of K-means: Differing Sizes

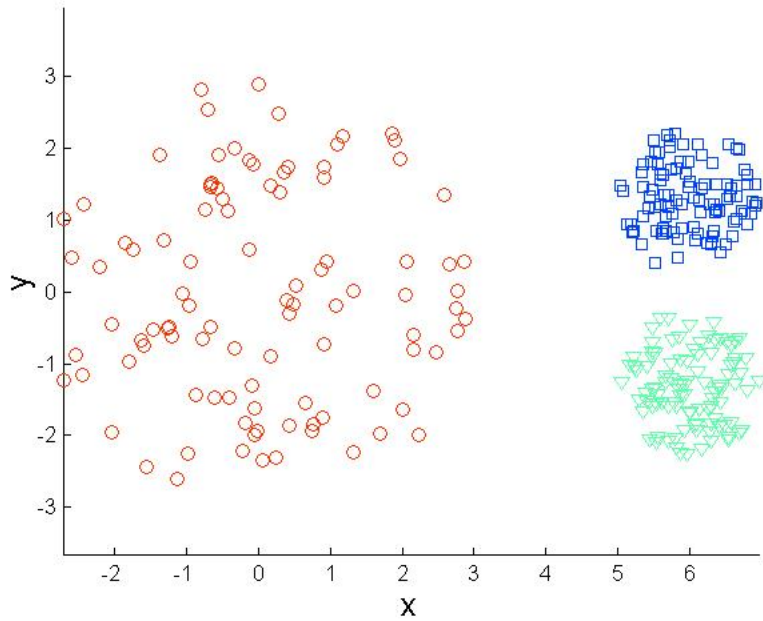


Original Points

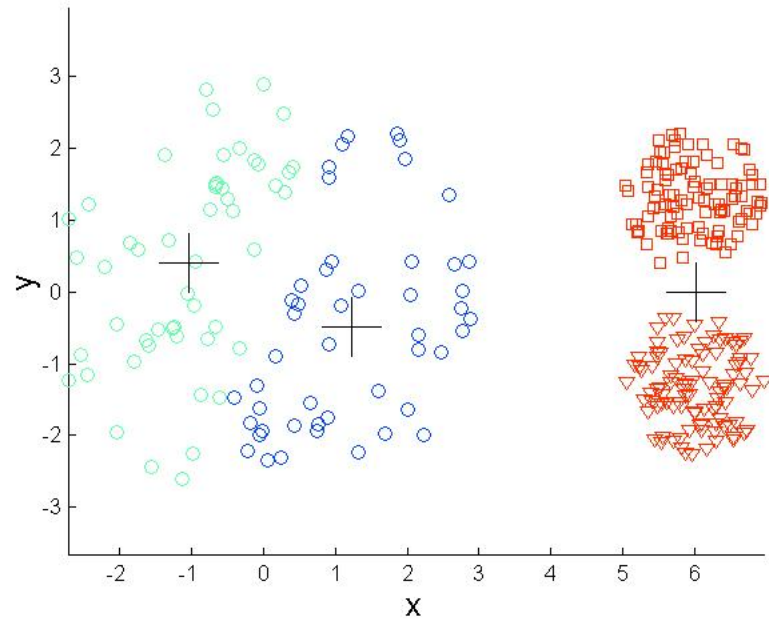


K-means (3 Clusters)

Limitations of K-means: Differing Density

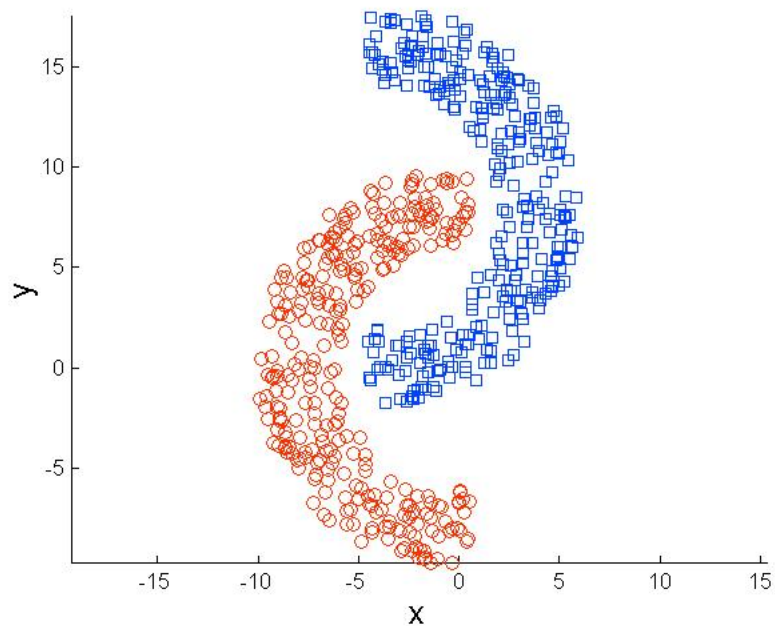


Original Points

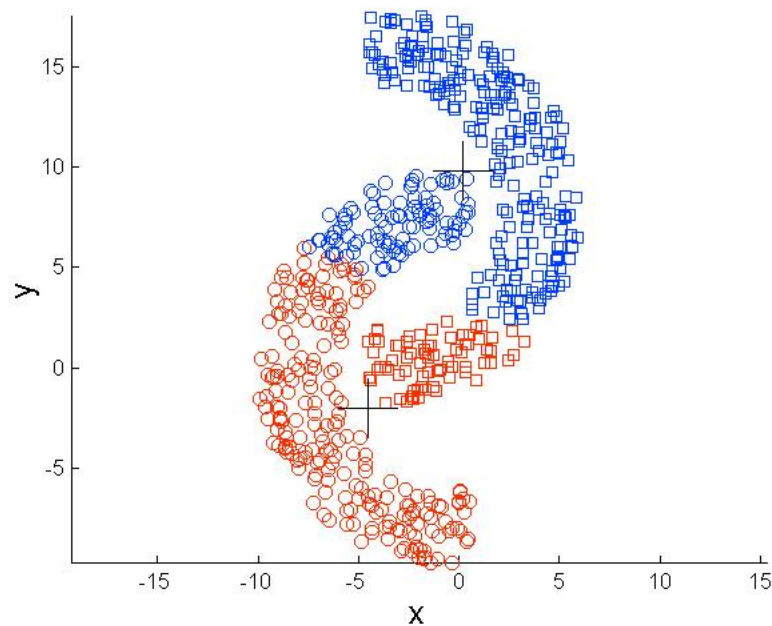


K-means (3 Clusters)

Limitations of K-means: Non-globular Shapes

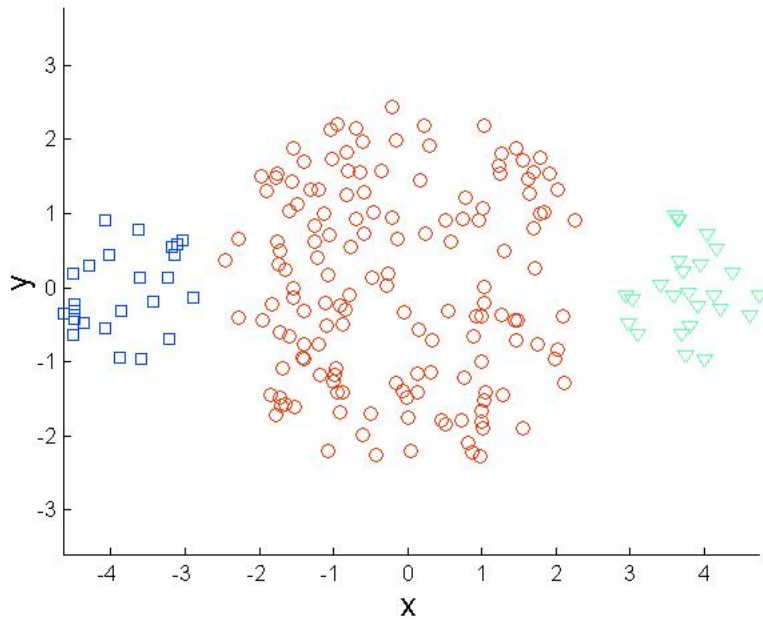


Original Points

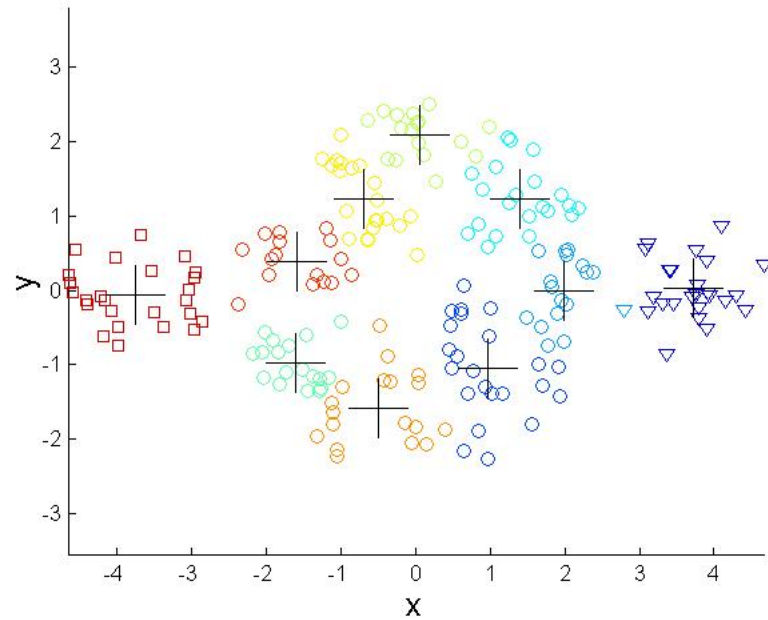


K-means (2 Clusters)

Overcoming K-means Limitations



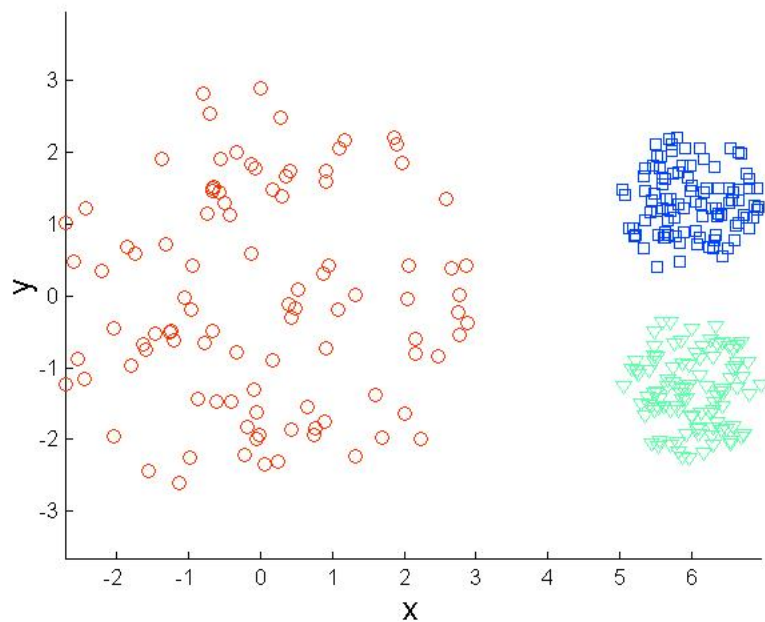
Original Points



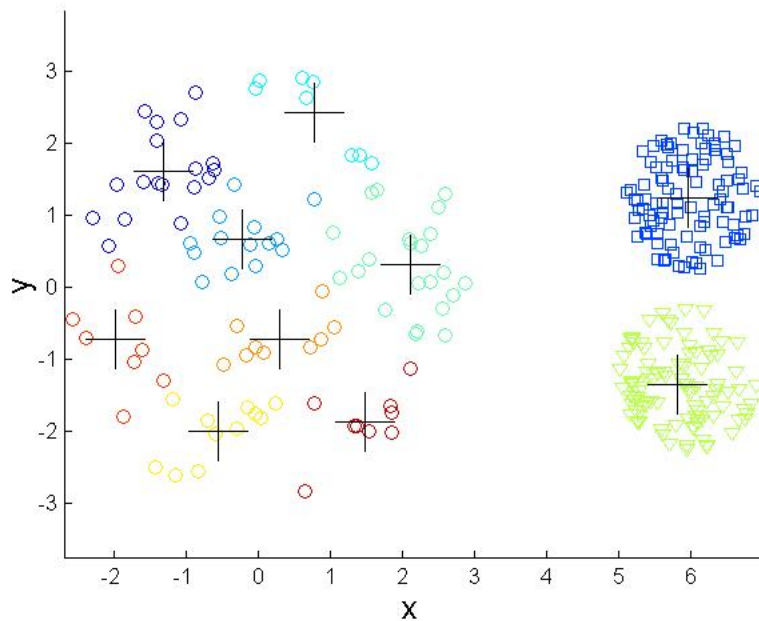
K-means Clusters

One solution is to use many clusters.
Find parts of clusters, but need to put together.

Overcoming K-means Limitations

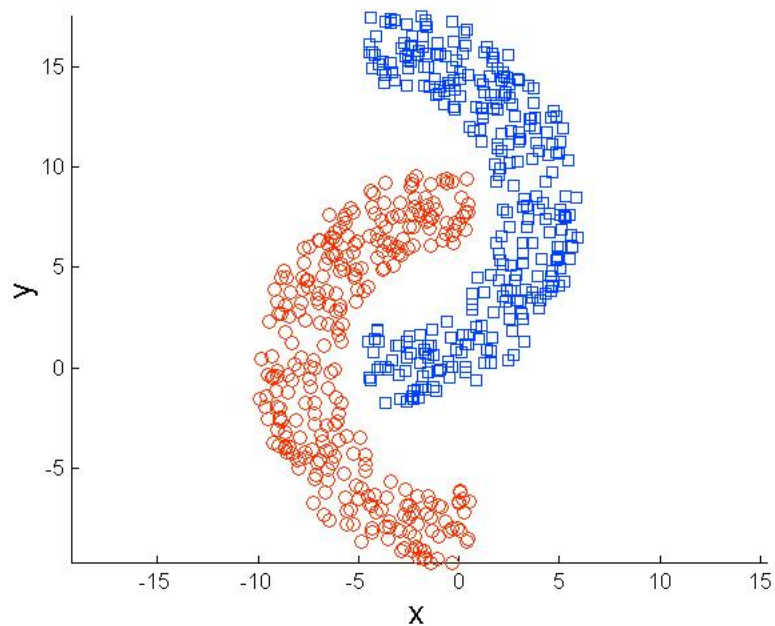


Original Points

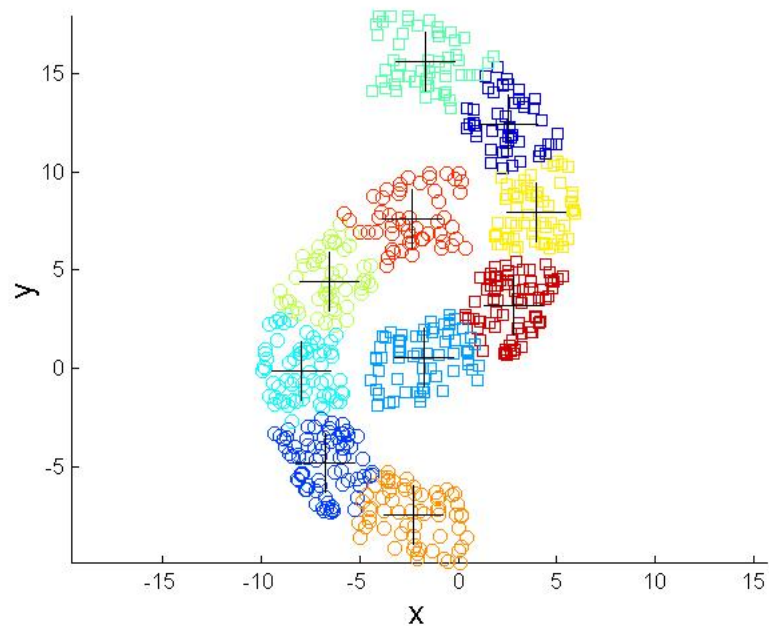


K-means Clusters

Overcoming K-means Limitations



Original Points



K-means Clusters

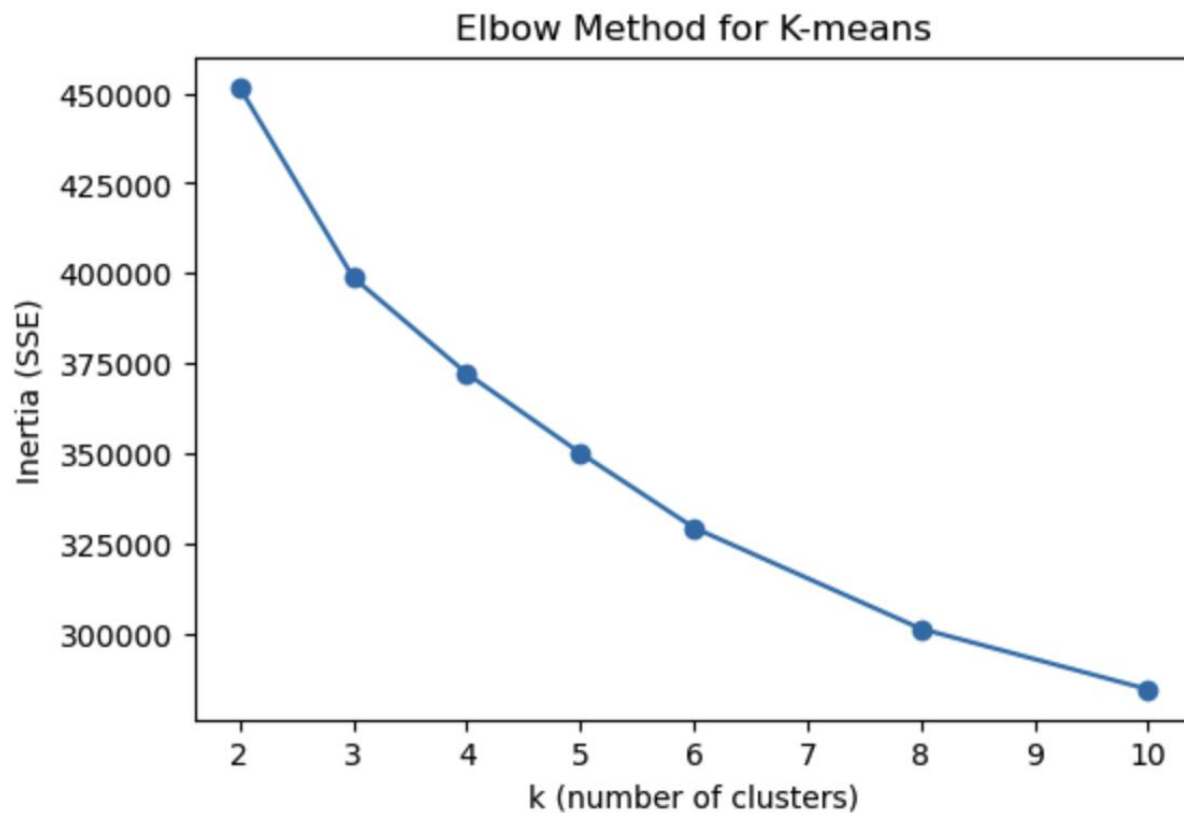
- What is cluster analysis
- Similarity and density measurement
- Types of cluster
- K-means and its variant model
- Case study

Case Study

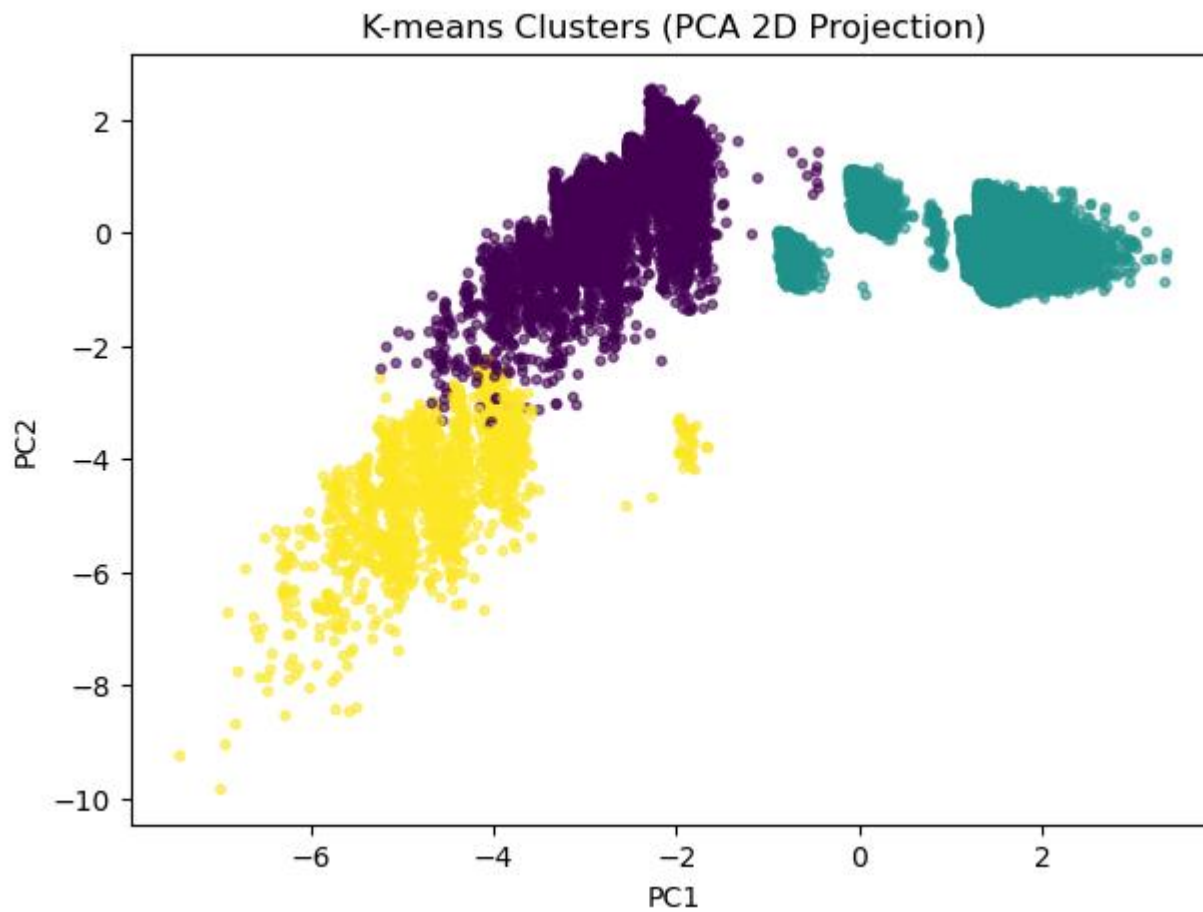
Bank Marketing Data:

- Bank client data:
 - 1 - age (numeric)
 - 2 - job : type of job (categorical: 'admin','blue-collar','entrepreneur','housemaid','management','retired','self-employed','services','student','technician','unemployed','unknown')
 - 3 - marital : marital status (categorical: 'divorced','married','single','unknown'; note: 'divorced' means divorced or widowed)
 - 4 - education (categorical: 'basic.4y','basic.6y','basic.9y','high.school','illiterate','professional.course','university.degree','unknown')
 - 5 - default: has credit in default? (categorical: 'no','yes','unknown')
 - 6 - housing: has housing loan? (categorical: 'no','yes','unknown')
 - 7 - loan: has personal loan? (categorical: 'no','yes','unknown')
- Related with the last contact of the current campaign:
 - 8 - contact: contact communication type (categorical: 'cellular','telephone')
 - 9 - month: last contact month of year (categorical: 'jan', 'feb', 'mar', ..., 'nov', 'dec')
 - 10 - day_of_week: last contact day of the week (categorical: 'mon','tue','wed','thu','fri')
 - 11 - duration: last contact duration, in seconds (numeric). Important note: this attribute highly affects the output target (e.g., if duration=0 then y='no'). Yet, the duration is not known before a call is performed. Also, after the end of the call y is obviously known. Thus, this input should only be included for benchmark purposes and should be discarded if the intention is to have a realistic predictive model.
- Other attributes:
 - 12 - campaign: number of contacts performed during this campaign and for this client (numeric, includes last contact)
 - 13 - pdays: number of days that passed by after the client was last contacted from a previous campaign (numeric; 999 means client was not previously contacted)
 - 14 - previous: number of contacts performed before this campaign and for this client (numeric)
 - 15 - poutcome: outcome of the previous marketing campaign (categorical: 'failure','nonexistent','success')
- Social and economic context attributes
 - 16 - emp.var.rate: employment variation rate - quarterly indicator (numeric)
 - 17 - cons.price.idx: consumer price index - monthly indicator (numeric)
 - 18 - cons.conf.idx: consumer confidence index - monthly indicator (numeric)
 - 19 - euribor3m: euribor 3 month rate - daily indicator (numeric)
 - 20 - nr.employed: number of employees - quarterly indicator (numeric)
- Desired target:
 - 21 - y - has the client subscribed a term deposit? (binary: 'yes','no')

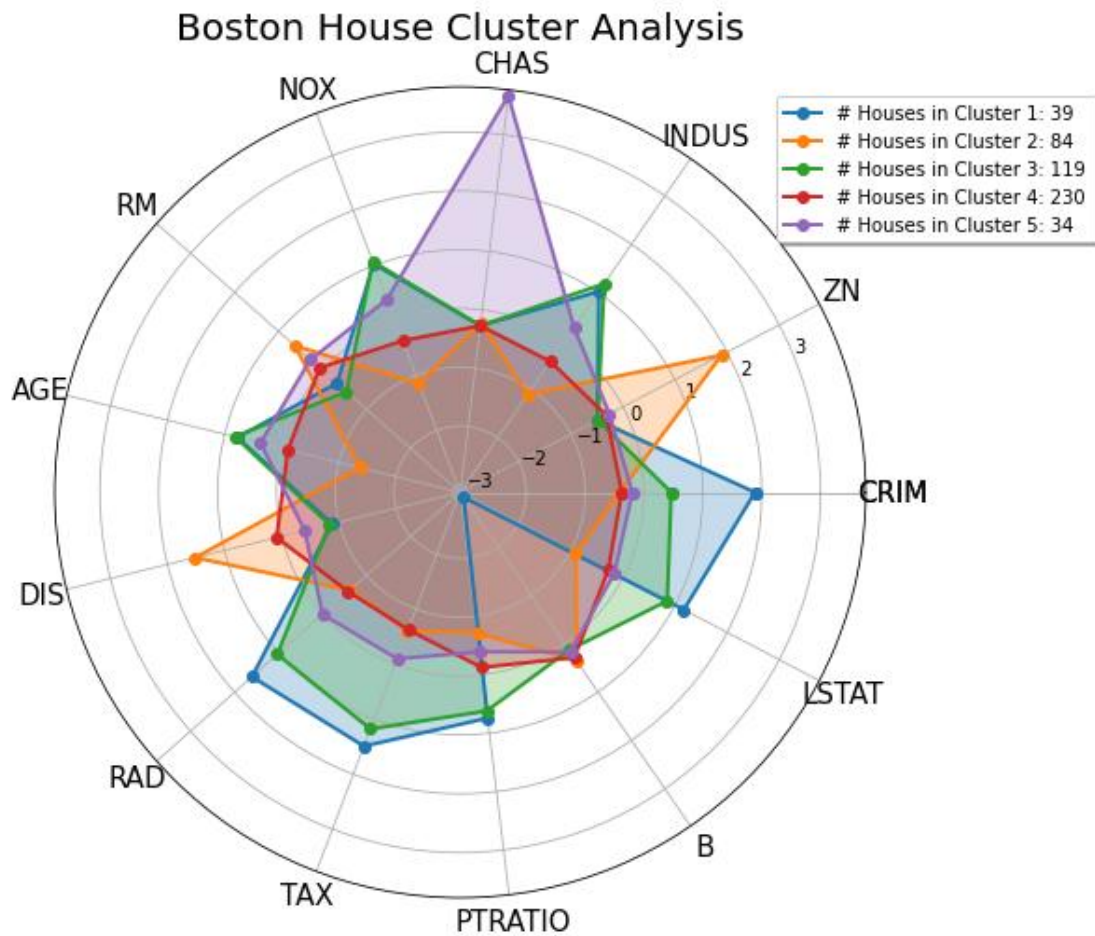
Case Study



Case Study



Case Study





Some Takeaways

- Cluster analysis
- Similarity and density
- K-means model



Acknowledgement

- Introduction to data mining (by Tan, Steinbach, Kumar)
- NYU: machine learning course (by Xi Chen)

The End

■ THANK YOU

Any questions?

